How-To Library

Informatica

Configuring Azure Virtual Network for Microsoft Azure Synapse SQL

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Abstract

You can use a Microsoft Azure Synapse SQL connection to connect to a Microsoft Azure Synapse SQL endpoint that resides in the Azure Virtual Network. This article explains how to configure a virtual network to create a secure and scalable, on-demand Azure infrastructure on Azure cloud.

Supported Versions

- Informatica Cloud® Data Integration Microsoft Azure Synapse SQL Connector

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Overview

Azure Virtual Network (VNet) enables many types of Azure resources, such as Azure Virtual Machines, to securely communicate with each other, the internet, and on-premises networks.

You can create a virtual network in the Azure portal to securely communicate with a Microsoft Azure Synapse SQL endpoint.

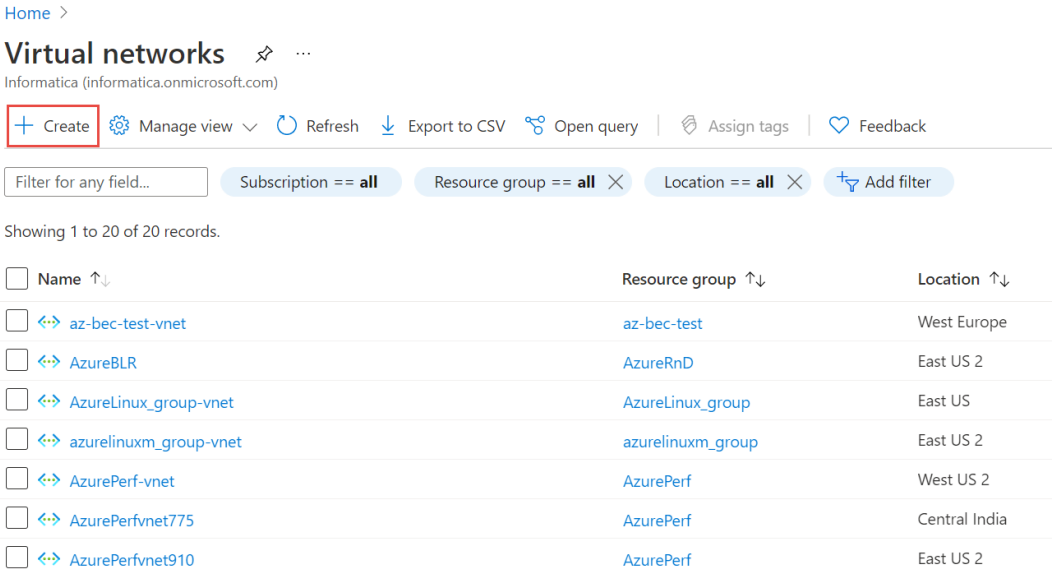
Before you connect to a Microsoft Azure Synapse SQL endpoint residing in a virtual network, perform the following prerequisite tasks:

1. Create a virtual network in Azure.
2. Add a subnet to the virtual network.
3. Create a virtual machine.
4. Create a virtual network rule.
5. Configure the Microsoft SQL Server database that hosts Microsoft Azure Synapse SQL to read from and write data to Azure storage.
6. Enable the virtual network for the Microsoft Azure Synapse SQL connection.

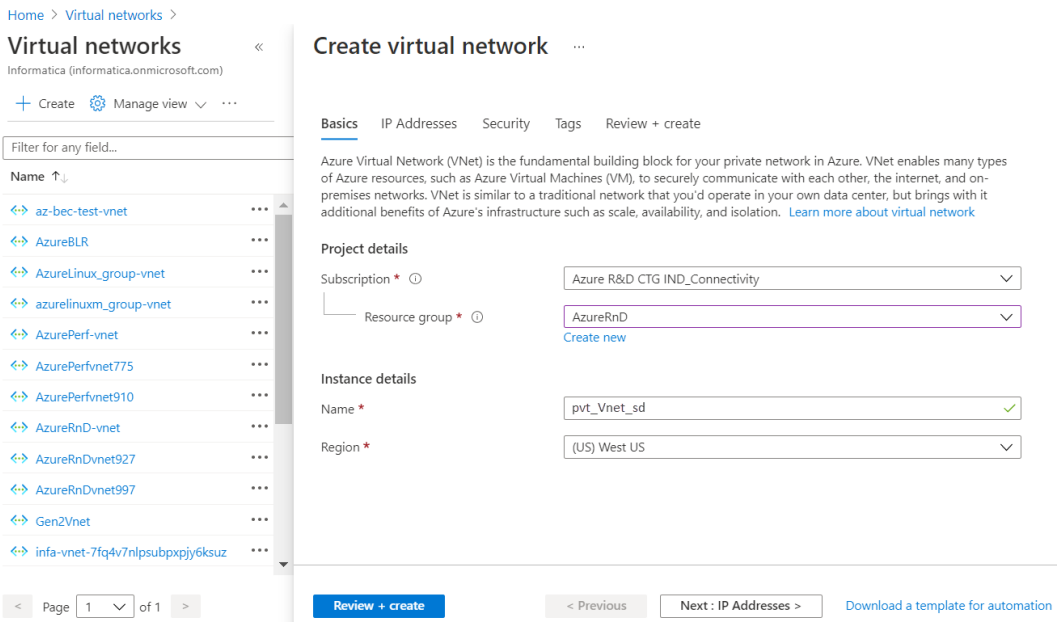
Creating a virtual network

Perform the following steps to create a virtual network in Azure:

1. Log in to the Azure portal
2. In the search box, enter **Virtual network**, and select **Virtual network** in the search results.
3. On the Virtual network page, click **Create**.



4. On the Basics tab, enter the project and instance details.



a. In the **Subscription** field, select your subscription for which you want to create the virtual network.

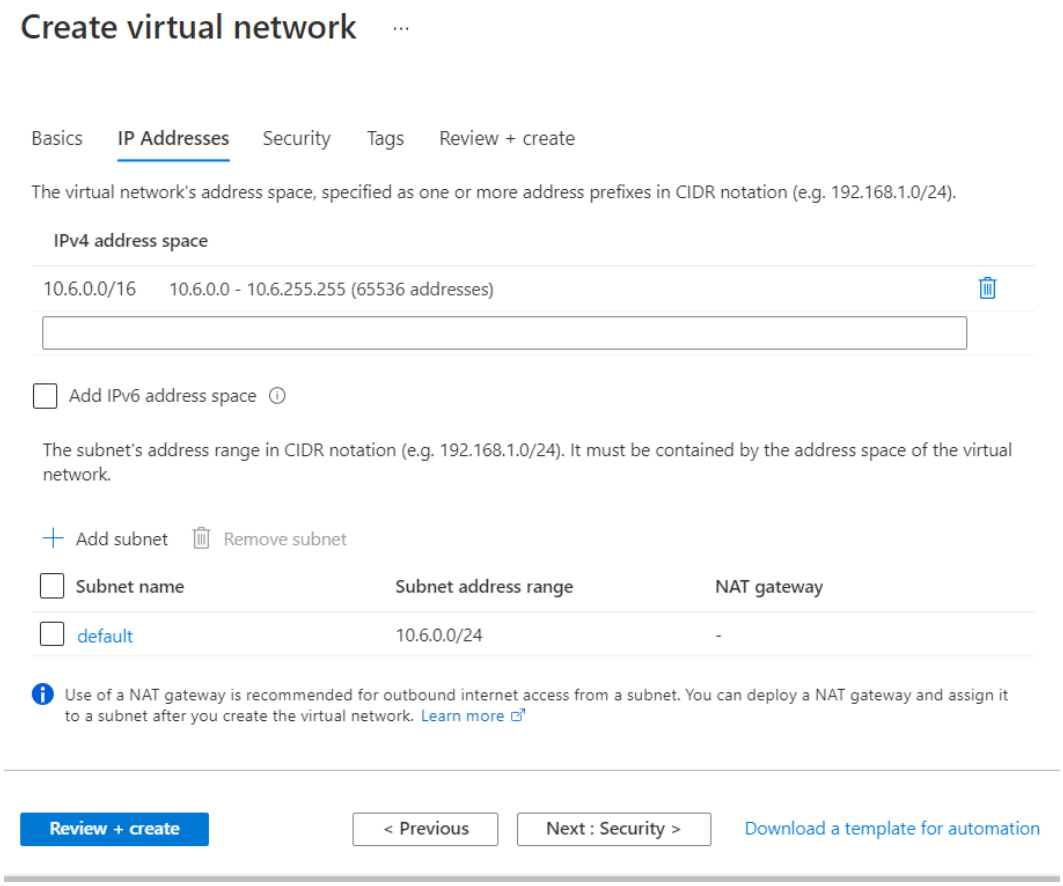
b. In the **Resource group** field, select the resource group in which the Azure resources are deployed and managed.

c. In the **Name** field, enter a name for the virtual network.

d. In the **Region** field, select the region

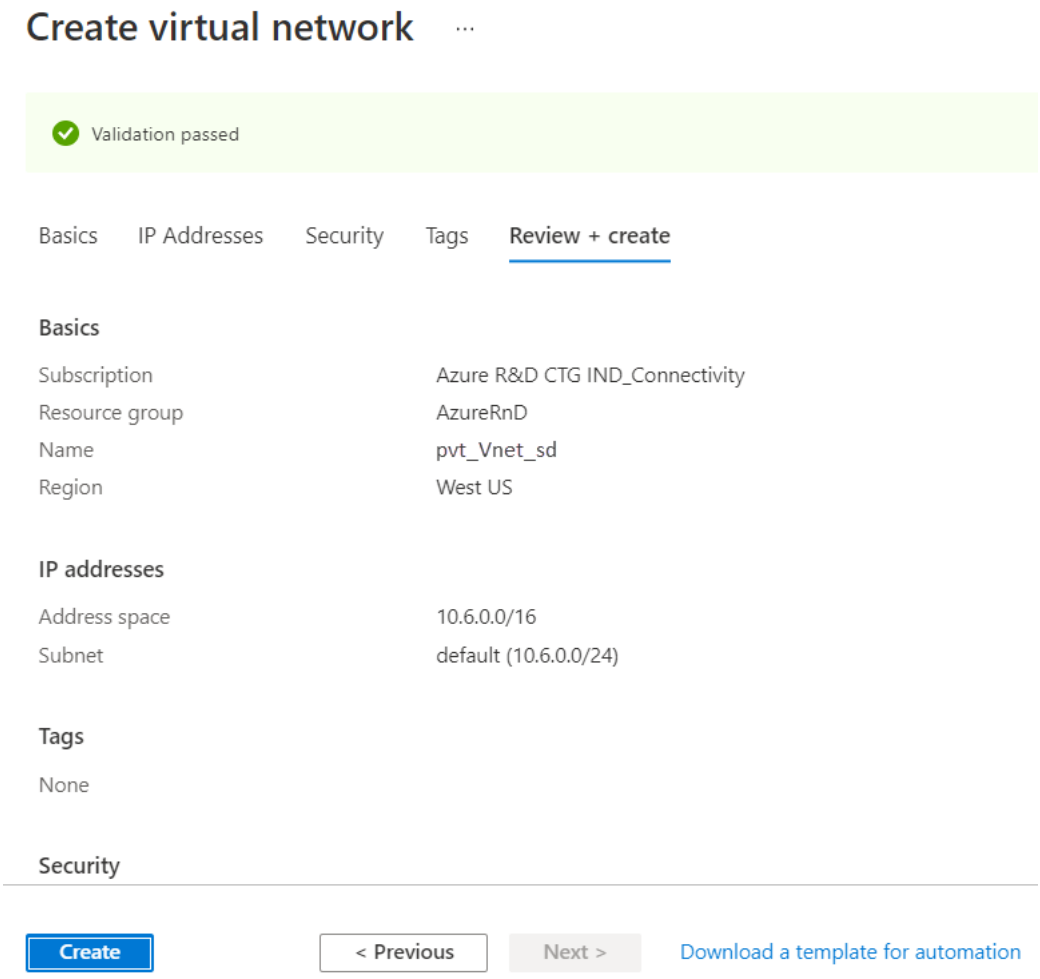
Note: Ensure that the virtual network and all the Azure resources are in the same region.

5. Click **Next: IP Addresses**.
The IP Addresses tab shows the IP address space of the virtual network and the address range of the subnet.



You can use the default subnet or add a new subnet. The subnet address range must be contained by the address space of the virtual network.

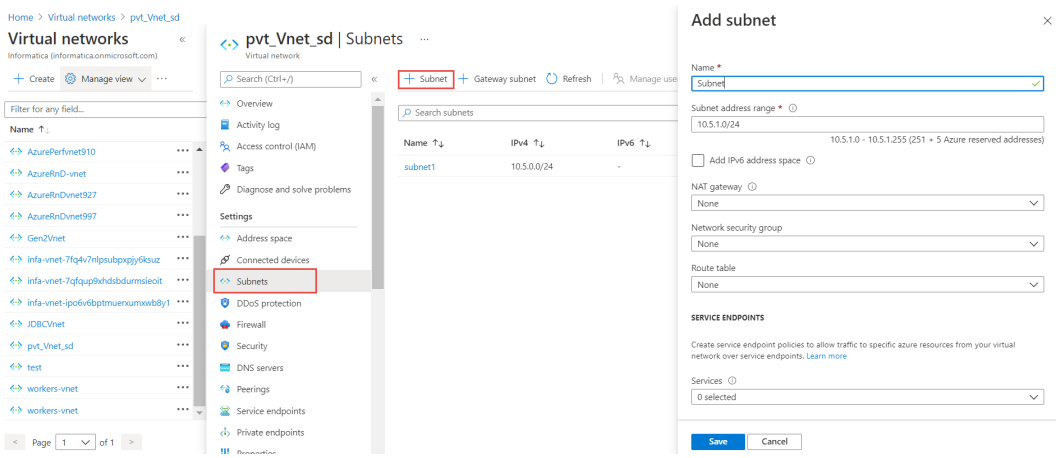
6. Click **Review + Create**, verify the configurations, and click **Create**.



Adding a subnet to the virtual network

A subnet is a range of IP addresses in the virtual network. You can segment the virtual network into one or more sub-networks and allocate a portion of the virtual network's address space to each subnet. You can then deploy Azure resources in a specific subnet.

1. Go to the virtual network that you created.
2. Under **Settings**, click **Subnets**.

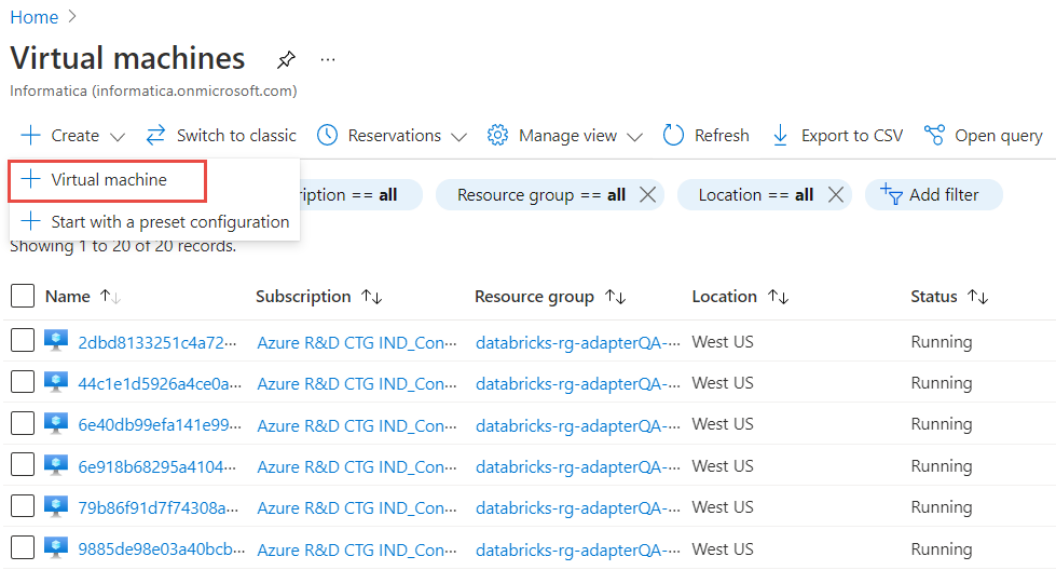


3. Click **Subnet**.
   The **Add subnet** page appears.
4. Enter a name for the subnet.
5. You can use the default subnet address range or specify an address range as per requirement.
   The subnet address range must be contained by the address space of the virtual network. The address range of a subnet which is in use cannot be edited.
6. Click **Save**.

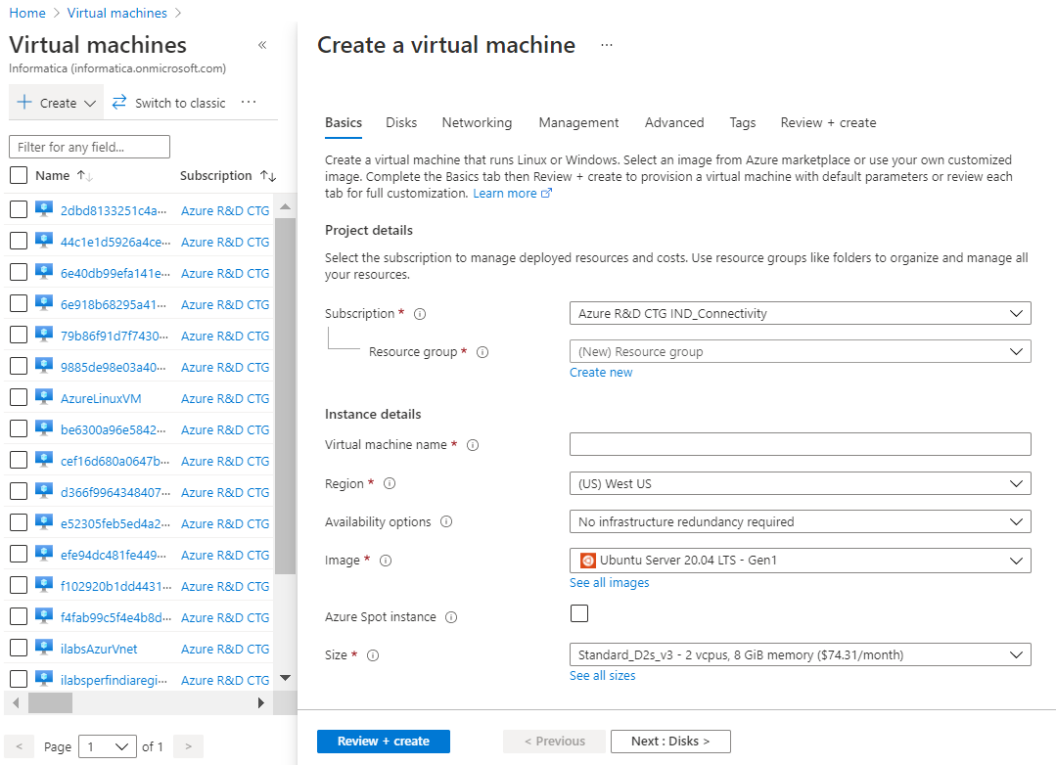
Creating a virtual machine in the subnet

Perform the following steps to create a virtual machine:

1. In the search box, enter **Virtual machines** , and select **Virtual machines** in the search results.
2. Click **Create** > **Virtual machine**.



3. On the **Basics** tab, enter the project, instance, and authentication details.



a. In the **Subscription** field, select the subscription for which you want to create the virtual machine.

b. In the **Resource group** field, select the resource group in which the Azure resources are deployed and managed.

c. In the **Virtual machine name** field, enter a name for the virtual machine.
The virtual machine name cannot be changed after the virtual machine is created.

d. In the **Region** field, select the region.

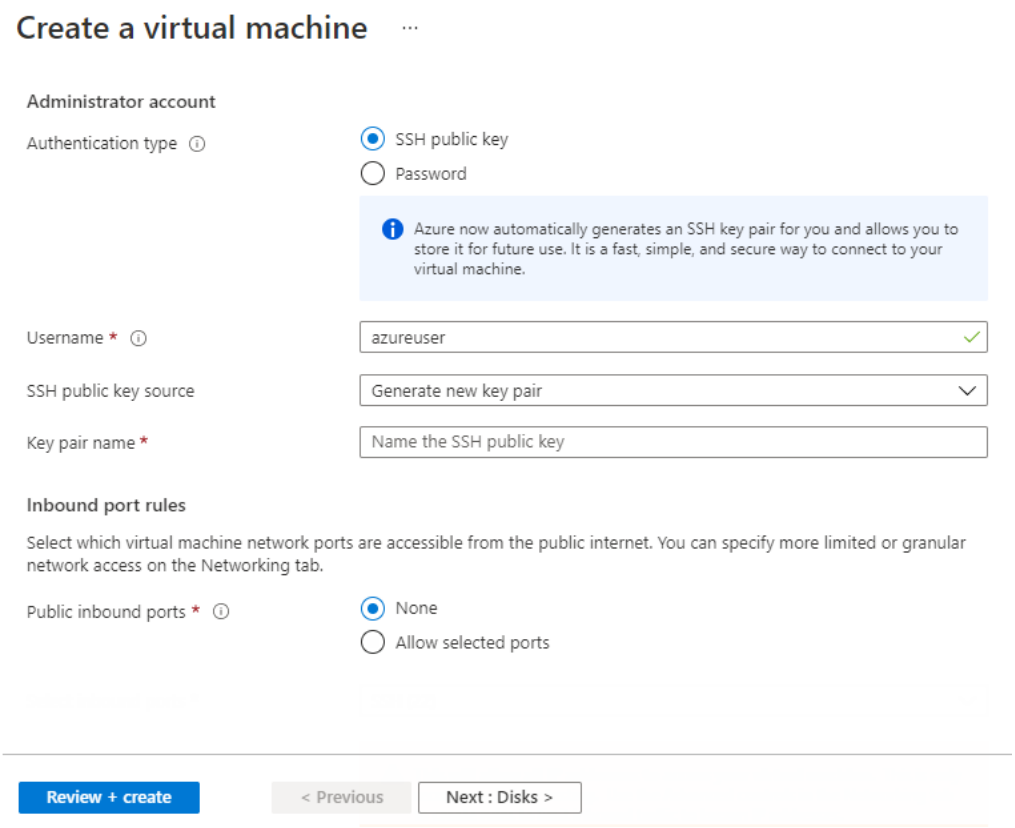
Note: Ensure that the subscription, resource group, and region for the virtual machine are the same as that of the virtual network.

e. In the **Availability** options field, you can choose to replicate the virtual machine in availability zones or availability sets to protect your applications and data from datacenter outages and maintenance events.

f. In the **Image** field, select the base operating system or application for the virtual machine.

g. In the **Size** field, select the size of the virtual machine that determines factors such as processing power, memory, and storage capacity.

h. In the Authentication type field, select if the administrator account will use username and password or SSH keys for authentication.

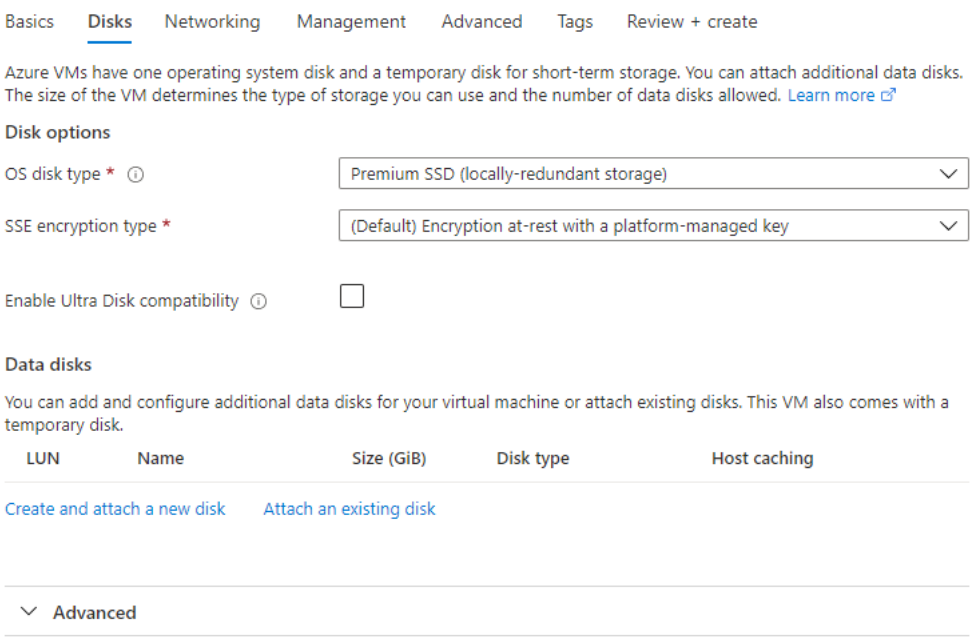


If you select the **SSH public key** option, enter the username and key pair name.

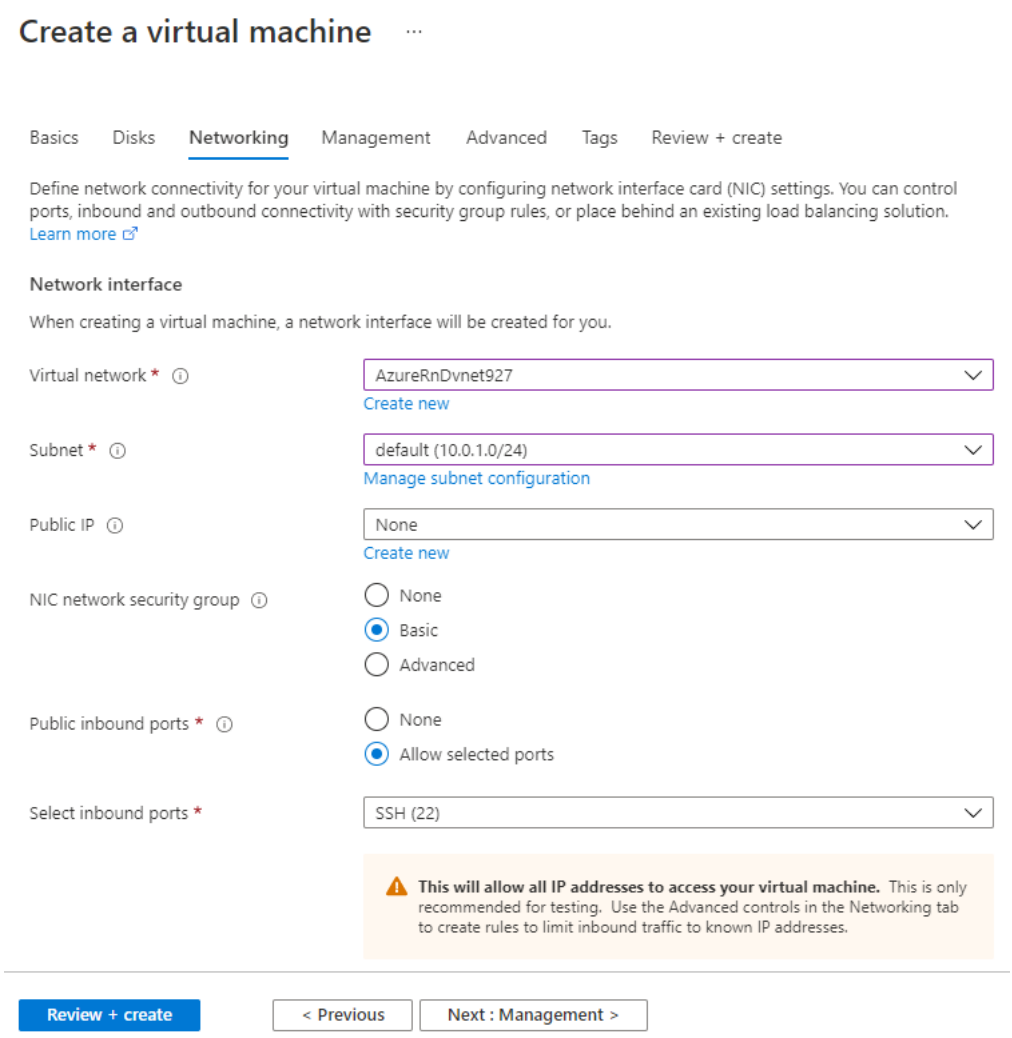
If you select the **Password** option, enter the username, password, and confirm password.

i. In the **Public inbound ports** field, select **None**.

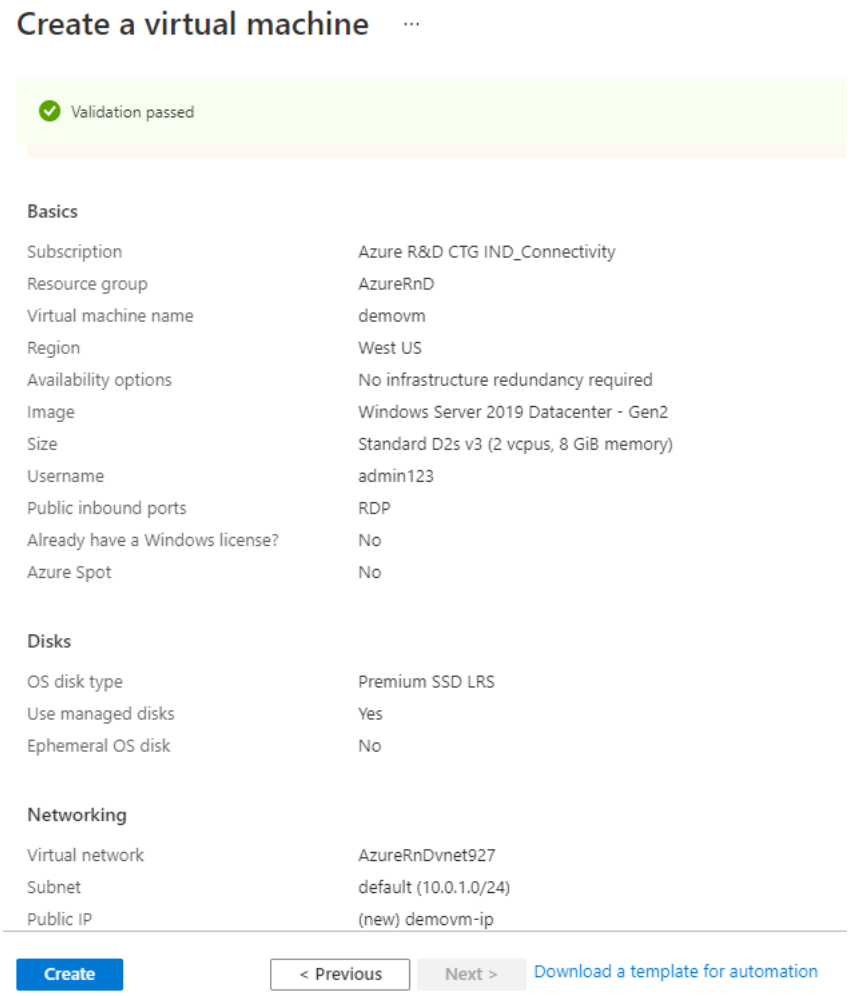
4. Click **Next : Disks**.
5. On the **Disks** tab, you can select the disk type for your virtual machine or use the default disk type. You can also configure additional data disks or attach existing disks.



6. Click **Next : Networking**.
7. On the **Networking** tab, select the virtual network and the subnet that you created.



8. Click **Review + create**.
9. On the **Review + create** tab, verify the configurations for the virtual machine.



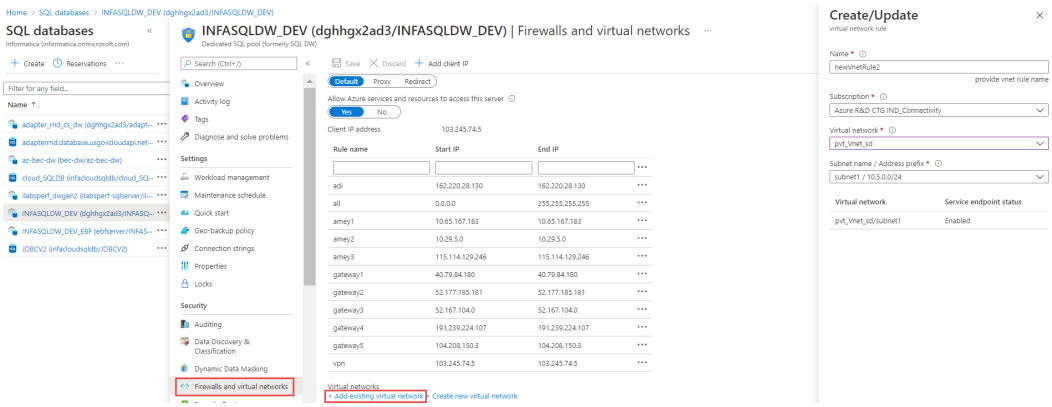
10. Click **Create**.

Creating a virtual network rule

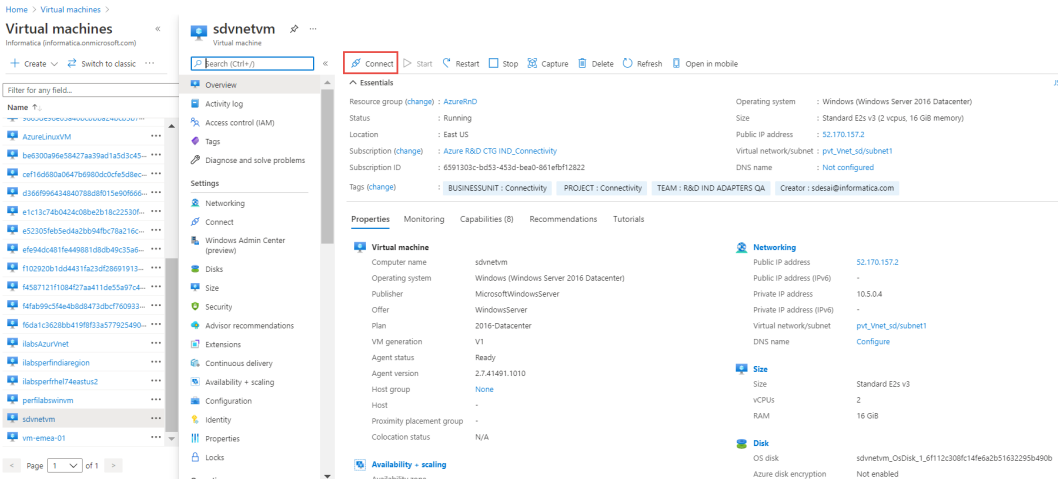
To allow the virtual network to access Microsoft Azure Synapse SQL, create a virtual network rule.

Perform the following steps to create a virtual network rule:

1. Go to your Microsoft Azure Synapse SQL account.
2. In the **Security** section, click **Firewalls and virtual networks**.
3. Under Virtual networks, click **Add existing virtual network**.



4. In the **Name** field, enter a name for the virtual network rule.
5. In the **Subscription** filed, select the subscription for the virtual network rule.
6. In the **Virtual network** field, select the virtual network that you created, to allow access to Microsoft Azure Synapse SQL.
7. In the **Subnet name** field, select the subnet.
8. Click **OK**.
9. Go to the virtual machine that you created in the subnet, and click **Connect**.



Configuring the Microsoft SQL Server database

You must configure the Microsoft SQL Server database that hosts Microsoft Azure Synapse SQL to read from and write data to Azure storage.

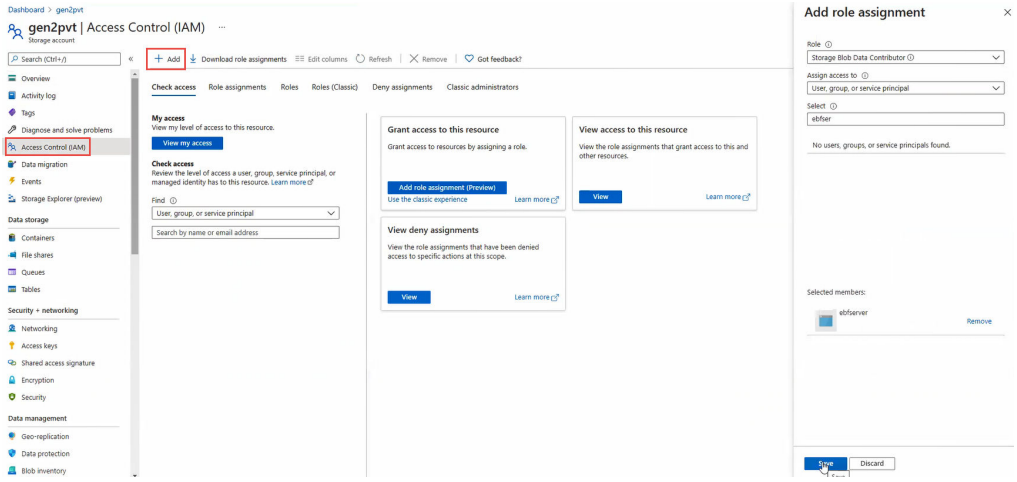
Ensure that the Azure storage account is a StorageV2 (general purpose v2) account.

Perform the following steps to configure the Microsoft SQL Server database:

1. Assign managed identity to the Microsoft SQL Server database to allow access to Azure storage.
   Open the Windows PowerShell command prompt window and run the following command:

   ```
   Connect-AzAccount Select-AzSubscription -SubscriptionId <your-subscriptionId> Set-AzSqlServer -ResourceGroupName <your-database-server-resourceGroup> -ServerName <your-database-servername> -AssignIdentity
   ```

2. Assign the Storage Blob Data Contributor role to the Microsoft SQL Server database to read from and write data to Azure storage.
    a. Log in to the Azure portal and go to the Azure storage account.



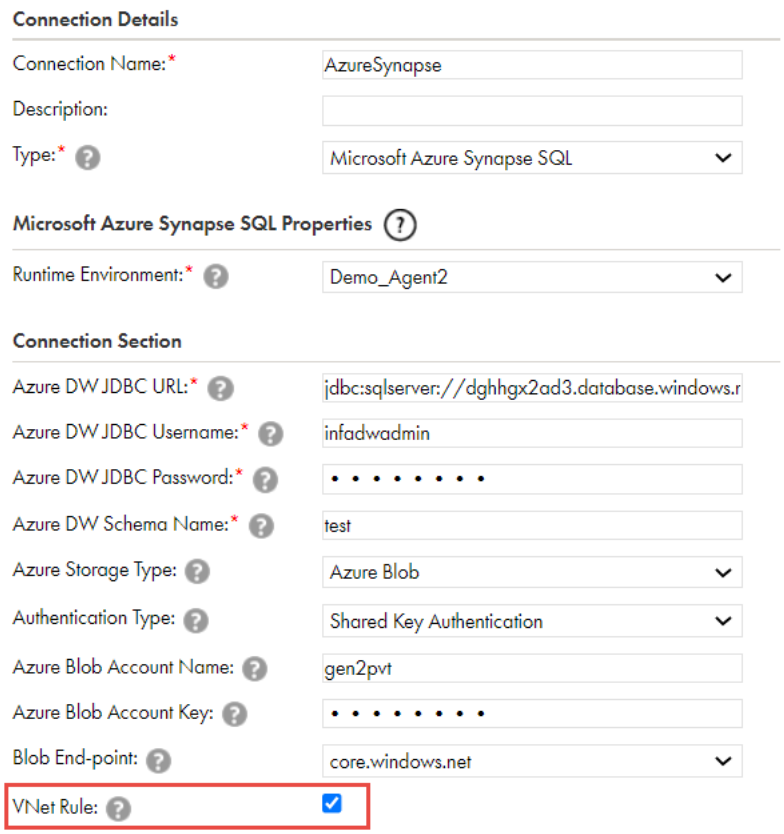
    b. Click **Add**.
    c. In the **Role** field, select **Storage Blob Data Contributor**.
    d. In the **Assign access to** field, select **User, group, or service principal**.
    e. In the **Select** field, select the Microsoft SQL Server database to which you want to assign the role.
    f. Click **Save**.
3. Configure the Azure Storage firewalls and virtual networks for the Microsoft SQL Server database.
    a. In the **Security + Networking** section of the Azure storage account, click **Networking**.
    b. On the **Firewalls and virtual networks** tab, add the virtual network that you created under **Virtual Networks**.
    c. Under **Resource instances**, add the Microsoft SQL Server database that hosts Microsoft Azure Synapse SQL to which you want to allow access to the Azure storage.

Enabling virtual network for Microsoft Azure Synapse SQL connection

After you complete the prerequisites tasks, enable the virtual network for Microsoft Azure Synapse SQL connection:

1. Log in to Informatica Intelligent Cloud Services.
2. Click **Administrator**.
3. Edit an existing connection or create a new connection.

4. Enable the virtual network in the connection properties.



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